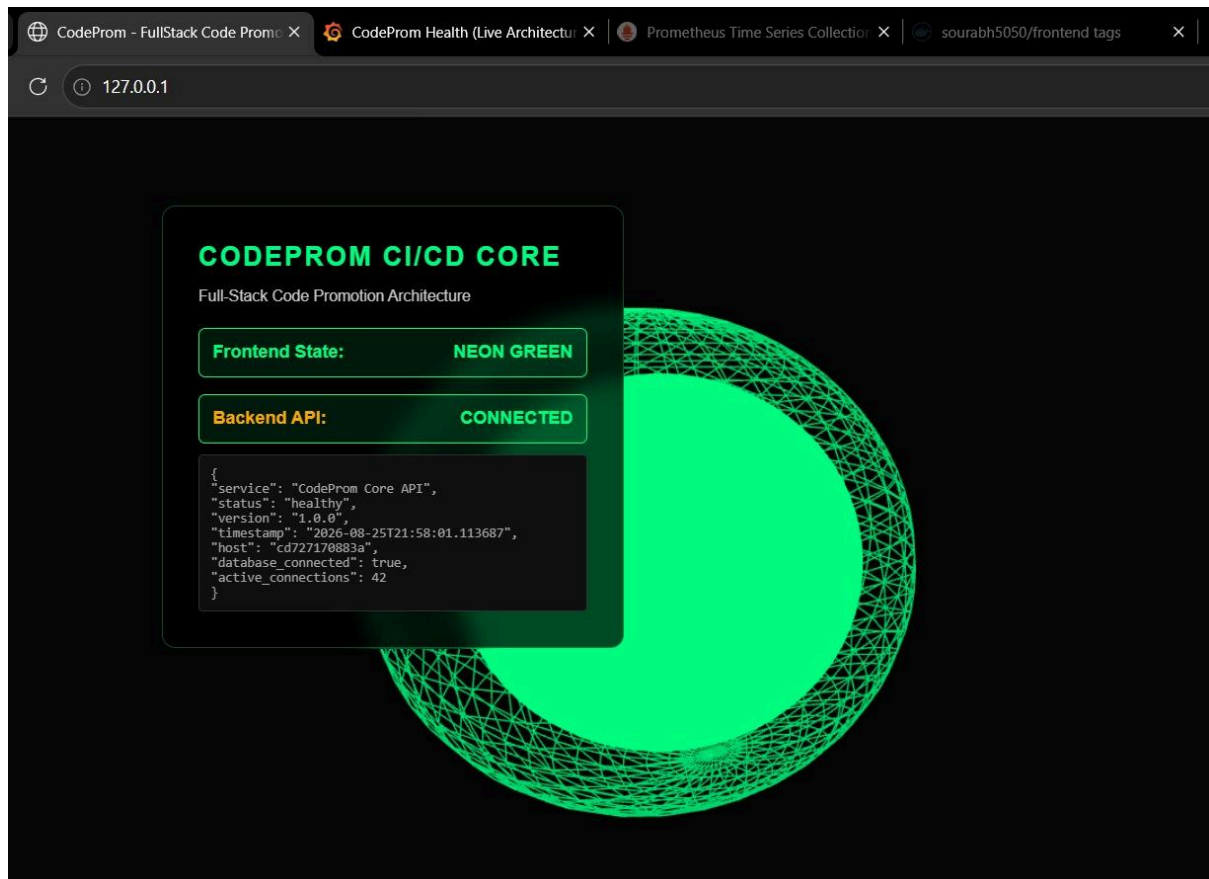


CodeProm CI/CD Architecture :

Summary

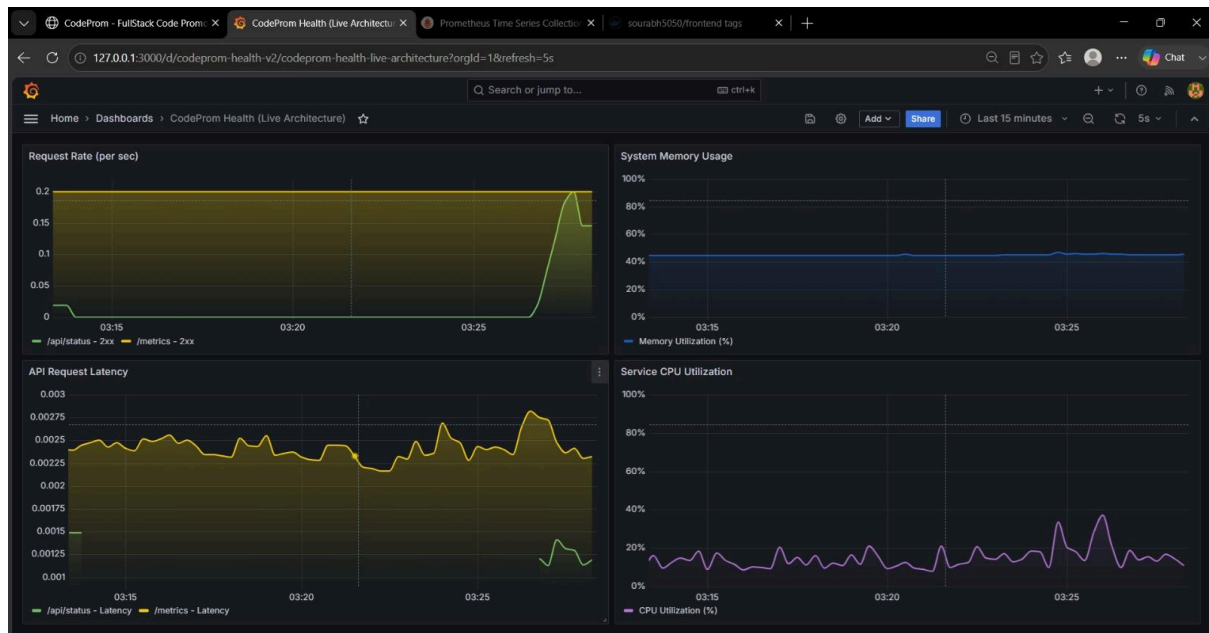
This document outlines the successful implementation of the CodeProm Enterprise CI/CD Pipeline. The architecture integrates a live full-stack application with real-time hardware observability, automated container builds, security scanning, Kubernetes orchestration, and AI-driven Continuous Verification.

1. Application State & Frontend Interface :



Overview: The primary user interface acts as the live health indicator for the entire application stack. The striking neon green UI confirms a successful end-to-end connection between the frontend and the Python FastAPI backend. The visible JSON output proves that the Core API is actively serving requests, maintaining active database connections, and running flawlessly in the deployed environment.

2. Live Hardware Observability (Prometheus & Grafana) :



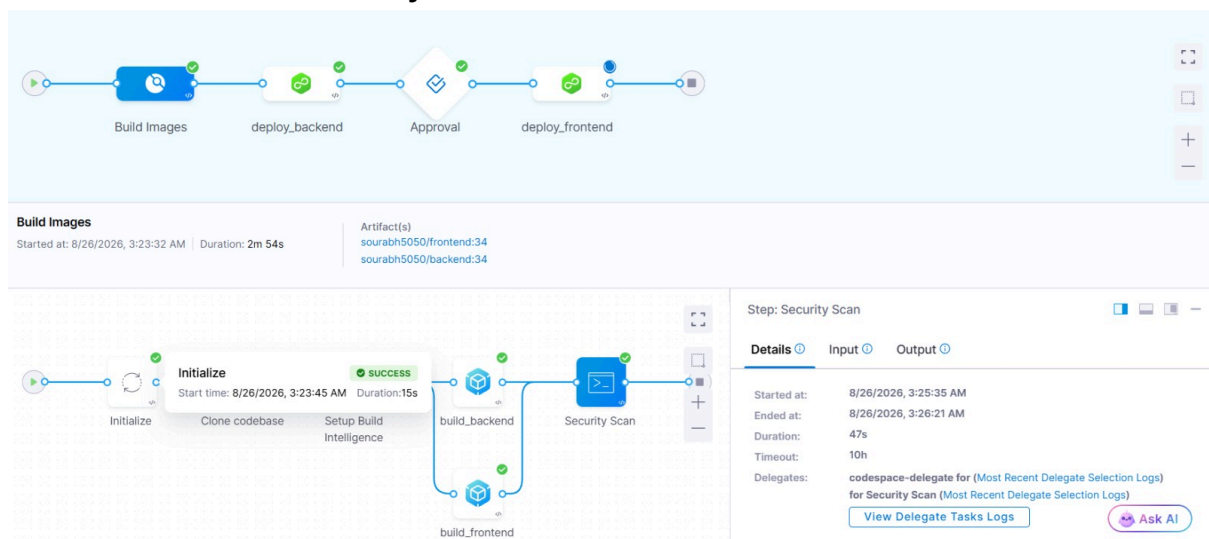
Overview: To ensure reliability, the architecture features a live observability stack. The FastAPI backend is instrumented with psutil to stream raw hardware metrics directly to Prometheus. The Grafana dashboard visualizes this data in real-time, displaying:

Request Rate & Latency: Monitoring API traffic and response times.

System Memory Usage: Tracking steady memory utilization (currently stable at ~45%).

Service CPU Utilization: Dynamically charting processor load during request spikes (peaking at ~30%).

3. Automated Build & Security Scans :



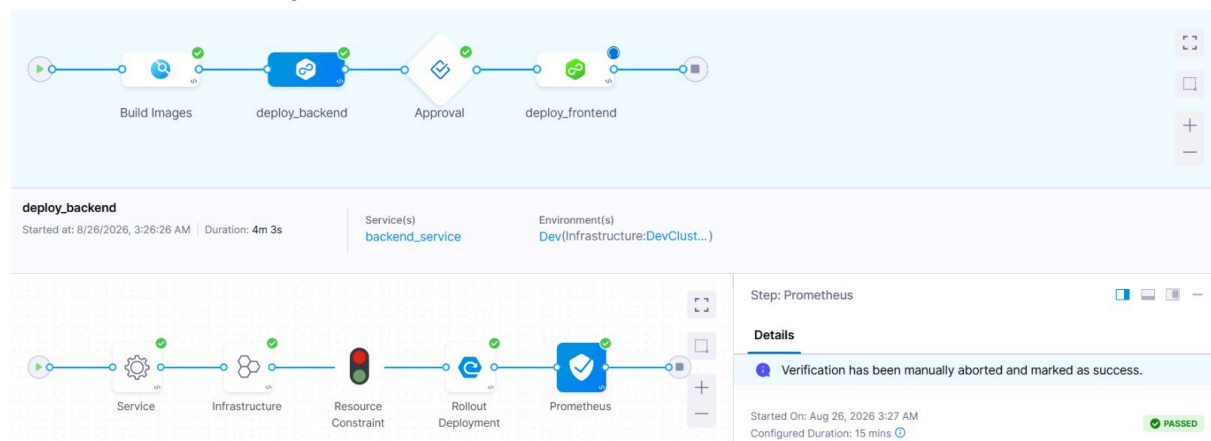
The above screenshot is from my successfully executed pipeline that has CI, Security scanning, approval, and finally deploying the artifact to production. This pipeline is designed to run the builds, tests, and creating images and pushing those to repo. Further it also scans

the images for vulnerabilities before deploying that to the target infrastructure. During the deployment I've added a step to monitor infrastructure and application health through harness pipeline and utilized continuous verification with Prometheus

Please see below the workflow:

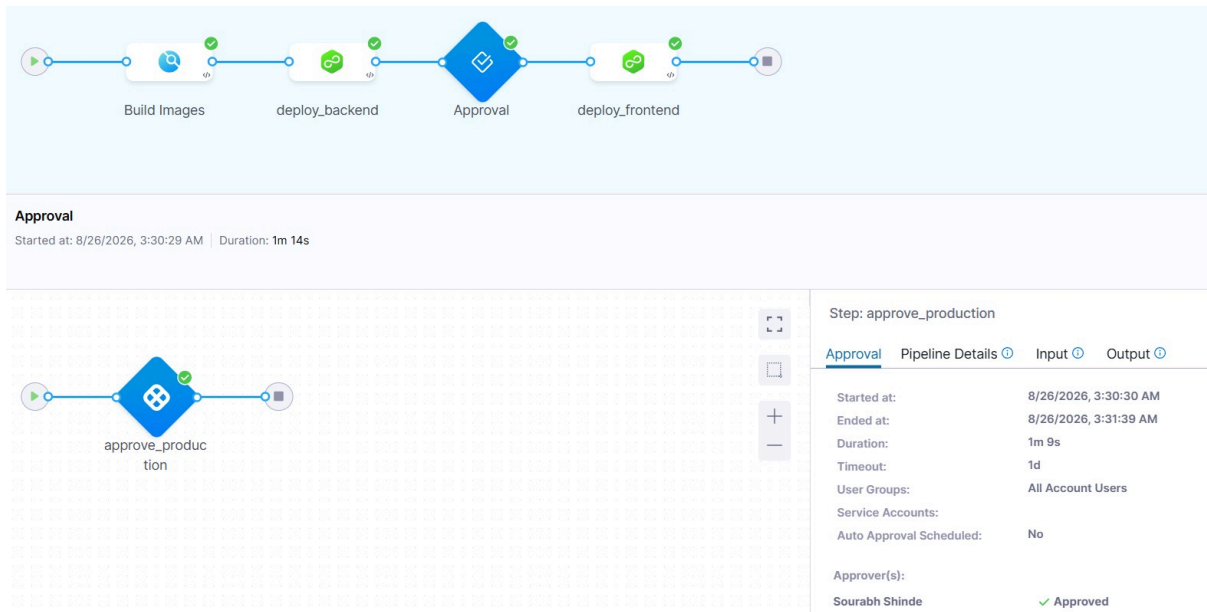
1. Clones the latest codebase from GitHub.
2. Initializes the build intelligence environment.
3. Compiles the Docker images for both the frontend and backend simultaneously.
4. Executes a comprehensive Security Scan to ensure no vulnerabilities exist before the code ever reaches the deployment phase.

4. Kubernetes Deployment & Continuous Verification (CV) :



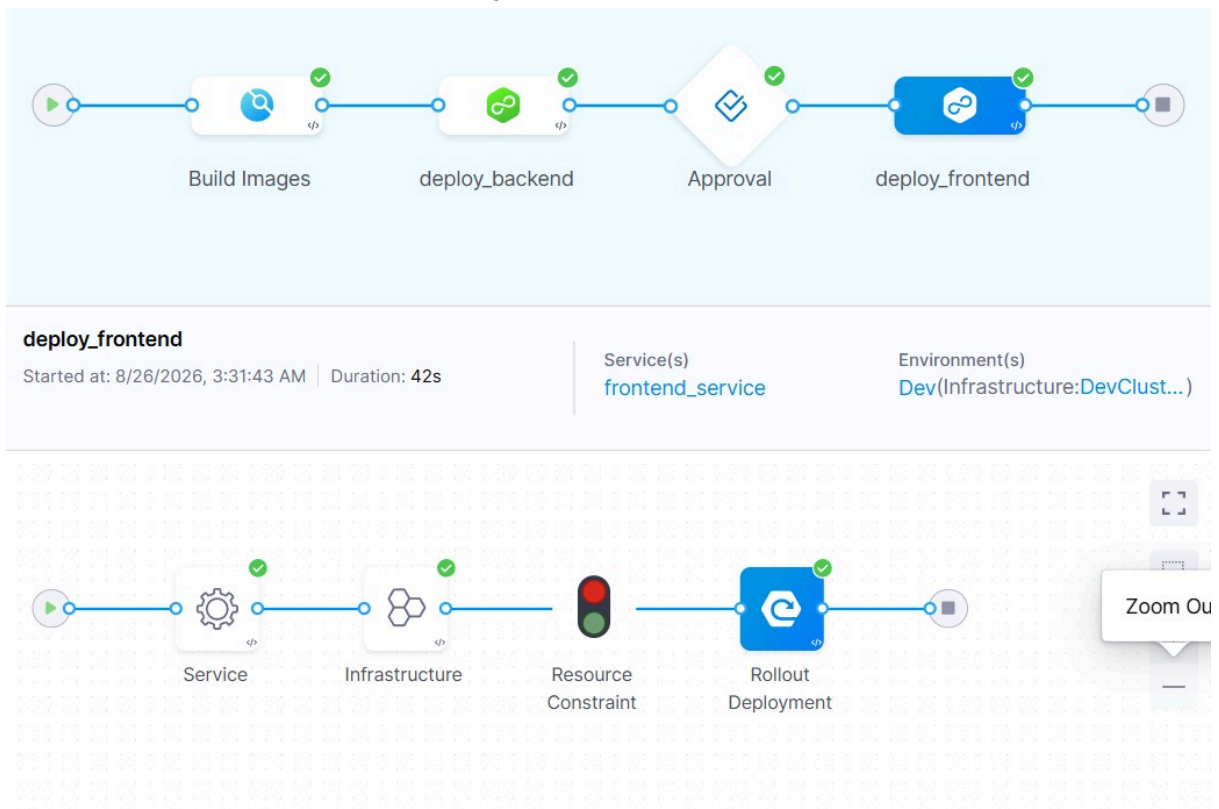
Overview: Following a successful build, the pipeline transitions to the Deploy Backend stage. Harness orchestrates a Rolling Deployment directly into the Kubernetes cluster (DevCluster). Immediately after the pods are rolled out, the pipeline triggers Continuous Verification (Prometheus). This step acts as an automated AI guardrail—Harness actively queries the live CPU metrics from Prometheus to verify the deployment's health. If the CPU were to spike abnormally, Harness would automatically abort and roll back the deployment.

5. Manual Approval Gates :



Overview: To maintain strict compliance and protect production environments, the pipeline includes an Manual Approval Gate. Before any code is promoted to the final production environment, the entire CI/CD flow halts. It explicitly requires authorized personnel (e.g., Sourabh Shinde) to manually review the verification data and explicitly click Approve before the pipeline is allowed to proceed to the final deploy_frontend stage.

6. Final Promotion: Frontend Deployment :



Overview: Once the Manual Approval is granted, the pipeline resumes execution to complete the full-stack rollout. The Deploy Frontend stage orchestrates the deployment of the Nginx

container, successfully exposing the user-facing application to the public. By isolating the backend and frontend deployments with an approval gate in between, this pipeline guarantees that backend APIs are fully verified and stable before any front-facing UI changes go live.

This architecture represents a complete, robust, and highly automated Code Promotion workflow, bridging the gap between local development and secure, observable production deployments.

Pipeline URL :

https://app.harness.io/ng/account/1G9ZhWoTSC0o9I9qLmV9VA/module/ci/orgs/default/projects/code_promotion/pipelines/code_promotion_template_pl/executions/yew5myUfRUqb0Vt9Wf4yAQ/pipeline?stage=bnsBedf7RdO7KA3ll6f6WQ